

Report 7: Grayscale Stretch

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1 Implementation

I used two CUDA reductions to find the minimum and maximum grayscale values. A 2D kernel then computes $(p - \min)255/(\max - \min)$. Constant images are kept unchanged to avoid division by zero. I checked the extrema and output against NumPy.

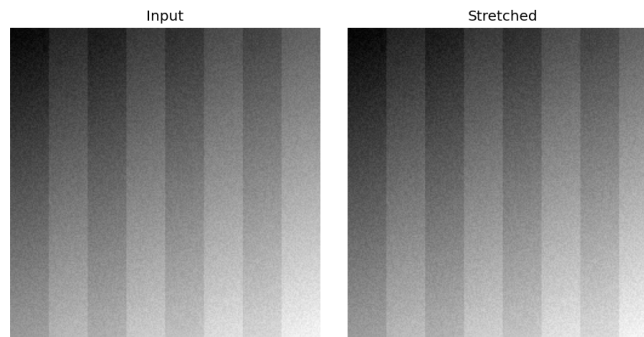
I ran the notebook on Google Colab with a Tesla T4 and a 256×256 sample image. GPU timings use `time.time()` after warm-up and synchronization, averaged over three runs; allocation and image transfers are excluded. Where measured, CPU time is one run; speedup is relative to that Python CPU implementation.

2 Results

Input minimum/maximum: 2/250; output: 0/255. Maximum pixel error: 0.

Reduction time: 0.9581 ms, including returning the two scalars to the host.

Block	Stretch kernel (ms)
(8, 8)	0.0718
(16, 16)	0.0548
(32, 16)	0.0558
(32, 32)	0.0481



3 Discussion

The output uses the full grayscale range. The 32×32 block gave the lowest map time (0.0481 ms). Finding the extrema took much longer than the pixel mapping on this small image.